

TUKTUT NOGAI, NT, Canada

WMO: 714920

Lat: 69.199N

Lon: 122.359W

Elev: 552

StdP: 94.87

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-37.4	-35.6	-41.5	0.1	-37.3	-39.5	0.1	-35.6	21.1	-16.2	19.3	-17.5	3.2	250	1.065

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	19.9	13.0	18.0	12.3	16.1	11.4	14.2	18.1	12.9	17.0	11.7	15.5	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	9.5	15.9	10.8	8.6	14.7	9.6	7.9	13.4	41.6	18.3	38.1	17.0	34.9	15.8	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.4	13.1	DB	-39.9	23.6	2.1	1.3	-41.4	24.6	-42.7	25.4	-43.8	26.1	-45.4	27.1	(4)
(5)			WB	-40.2	15.8	2.1	1.4	-41.7	16.9	-42.9	17.7	-44.1	18.5	-45.6	19.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.4	-23.8	-23.9	-22.8	-14.1	-4.4	5.6	9.9	6.5	0.8	-8.5	-17.7	-21.8	(6)
(7)		DBStd	13.62	6.66	6.50	7.97	6.51	5.88	5.17	4.56	4.43	5.19	4.37	5.24	6.20	(7)
(8)		HDD10.0	7191	1047	948	1016	723	448	151	61	127	278	574	832	985	(8)
(9)		HDD18.3	10135	1305	1182	1274	973	706	383	262	366	527	833	1082	1243	(9)
(10)		CDD10.0	98	0	0	0	0	1	18	57	20	2	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.0	4.9	4.6	5.0	5.3	5.2	5.0	4.9	5.5	6.2	5.1	4.9	(14)
(15)	Precipitation	PrecAvg	200	9	8	9	8	11	18	32	38	26	19	13	10	(15)
(16)		PrecMax	262	27	24	31	25	29	38	56	75	53	38	28	22	(16)
(17)		PrecMin	143	2	1	2	1	2	6	14	18	9	5	4	3	(17)
(18)		PrecStd	34	4	5	6	5	6	7	9	14	9	9	7	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.1	-7.5	-3.5	5.4	14.6	21.2	22.4	21.4	14.6	6.2	-4.9	-5.7	(19)
(20)			MCWB	-3.9	-8.5	-4.6	2.9	10.1	13.0	15.2	14.5	10.8	4.8	-5.3	-6.2	(20)
(21)		2%	DB	-7.2	-9.4	-5.8	1.0	8.6	18.3	20.5	18.2	11.7	0.8	-7.0	-8.8	(21)
(22)			MCWB	-8.0	-10.1	-6.7	-0.7	4.7	11.7	13.5	12.9	8.9	0.3	-7.3	-9.2	(22)
(23)		5%	DB	-10.0	-11.5	-7.7	-1.9	5.7	15.8	18.9	15.7	9.9	-1.2	-8.9	-10.8	(23)
(24)			MCWB	-10.5	-12.0	-8.4	-3.1	2.7	10.6	12.8	11.6	7.5	-1.8	-9.5	-11.2	(24)
(25)		10%	DB	-14.4	-13.9	-10.5	-4.3	3.6	13.4	17.2	13.5	7.9	-2.6	-10.4	-12.8	(25)
(26)			MCWB	-14.9	-14.4	-11.1	-5.2	1.3	9.0	12.1	10.1	6.3	-3.2	-10.8	-13.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.0	-8.3	-4.4	3.1	9.6	13.8	16.1	15.5	11.3	4.7	-5.0	-6.5	(27)
(28)			MADB	-3.1	-7.7	-3.6	5.6	14.9	20.1	21.0	20.0	13.5	6.2	-4.9	-5.9	(28)
(29)		2%	WB	-7.9	-10.0	-6.6	-0.6	5.2	12.3	14.7	13.4	9.2	0.4	-7.3	-9.1	(29)
(30)			MADB	-7.3	-9.5	-5.8	0.9	8.0	17.5	18.8	16.7	11.5	0.8	-7.1	-8.8	(30)
(31)		5%	WB	-10.5	-12.0	-8.4	-3.1	3.0	10.9	13.6	11.8	7.7	-1.6	-9.4	-11.2	(31)
(32)			MADB	-9.9	-11.5	-7.7	-1.9	5.3	15.1	17.4	15.5	9.6	-1.3	-9.0	-10.9	(32)
(33)		10%	WB	-14.8	-14.4	-11.1	-5.2	1.4	9.4	12.5	10.5	6.3	-3.1	-10.8	-13.2	(33)
(34)			MADB	-14.4	-13.9	-10.5	-4.3	3.4	13.2	16.4	13.1	7.7	-2.7	-10.4	-12.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	7.9	8.5	7.6	7.3	8.1	6.4	5.1	4.8	5.8	6.5	(35)
(36)			MCDBR	9.4	8.8	8.4	8.6	8.3	10.2	9.7	8.8	7.1	5.1	6.7	7.9	(36)
(37)			MCWBR	9.0	8.5	7.9	7.3	5.9	5.6	5.0	4.7	4.7	4.9	6.5	7.6	(37)
(38)		5% WB	MCDBR	9.5	8.9	8.4	7.9	7.8	9.5	9.0	8.2	7.0	4.9	6.7	7.8	(38)
(39)			MCWBR	9.1	8.5	7.9	6.8	5.7	5.4	5.2	5.0	4.8	4.7	6.6	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.148	0.200	0.236	0.283	0.319	0.319	0.333	0.315	0.278	0.213	0.145	N/A	(40)	
(41)		taud	1.902	2.077	2.235	2.162	2.177	2.417	2.422	2.488	2.491	2.138	1.914	N/A	(41)	
(42)		Ebn at Noon	70	594	813	857	861	879	850	828	769	552	55	0	(42)	
(43)		Edh at Noon	4	41	70	104	117	96	92	77	57	36	3	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.02	0.46	2.07	4.56	6.24	6.14	5.17	3.32	1.72	0.52	0.07	0.00	(44)	
(45)		RadStd	0.00	0.05	0.09	0.19	0.42	0.45	0.43	0.39	0.19	0.06	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (5 neighbors)		+0.56	+1.18	+1.54	N/A	N/A	N/A	-221	-215	N/A	N/A		

Nomenclature: See separate page